1.

(e)

For Part (e), I compared the models from parts (b), (c), and (d). The smallest model was the int8 quantized model from part (b), which had a TFLite size of 7.91 KB and still achieved 100% test accuracy. The float16 and dynamic quantized models were slightly larger at 8.77 KB and 8.44 KB, respectively, while the base model was 14.01 KB, the pruned model was 13.36 KB, and the distillation model was 29.50 KB. Based on these results, the int8 model already gave the best size–performance tradeoff among the methods tested.

I don't think a large additional size reduction is possible without sacrificing performance. Pruning did not help in this case, because the pruned model remained relatively large and its test accuracy dropped sharply to 0.17. Distillation also did not reduce the model size, since the student model was actually larger than the int8-compressed baseline. This suggests that the main source of compression was quantization, and the model is already near the practical limit for this assignment.

If I were to compress the model even further, the best approach would be to first compress the neural network architecture and then do int8 quantization. This can be done by reducing the number of hidden layers or the number of hidden units, which could lead to a decrease in accuracy. However, since the model after int8 quantization is still able to achieve perfect results, there is no need to compress it further.

2.

1)
The learning blocks that you can use include blocks for classification, regression, anomaly detection using K-means, anomaly detection using Gaussian Mixture Models, visual anomaly detection, and transfer learning blocks for image/object/audio classification.
For instance, a classification learning block may be used to develop a multiclass classifier using Keras, a regression block can be used to make predictions involving continuous variables, while anomaly detection can be carried out using either K-means or Gaussian Mixture Models.
A visual anomaly detection block can be used to detect anomalies in images, and transfer learning blocks may be used for image or audio classification.

2)

For a typical classifier, the common layers are Dense layers for feature transformation, Dropout for regularization, and an output Dense + softmax layer for class prediction.

3)

The free-tier Deployment section includes options such as Deploy as a customizable library, Deploy as a pre-built firmware, Deploy as a Linux .eim binary, Deploy as a Docker container, and Deploy in your web browser.

4)

Edge Impulse supports synthetic data generation for images, speech, and audio. For images, it uses DALL-E,  for speech or keyword spotting, it uses Whisper, and for audio events, it uses ElevenLabs sound generation. Synthetic data is useful because it can expand datasets when real data is limited, expensive, unbalanced, or hard to collect. It also helps models see rare or difficult scenarios that may not appear often in the real world.